EO M270.01 – IDENTIFY ASPECTS OF AIRCRAFT MANUFACTURING

Teaching Points

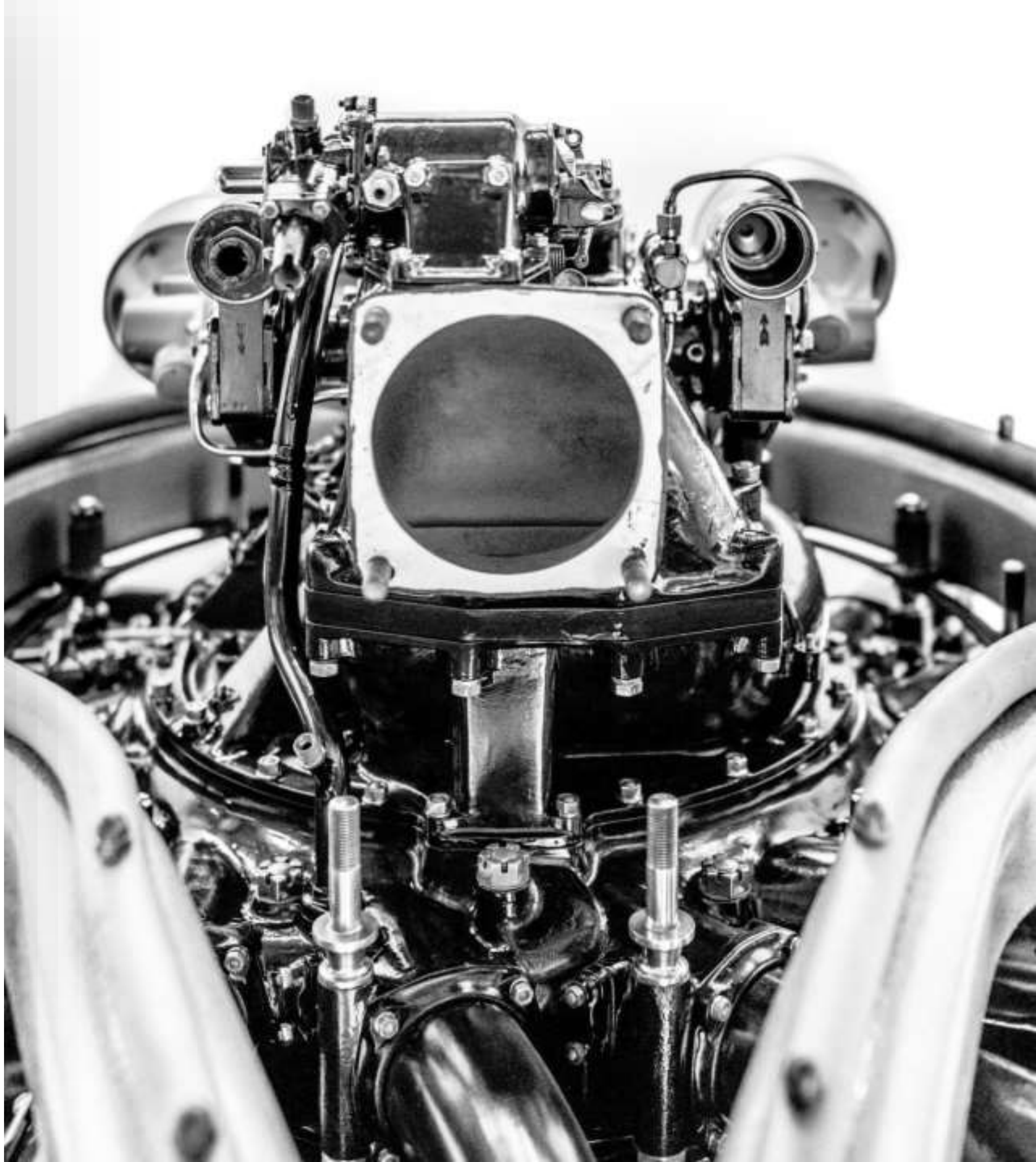
TP1 Identify Aircraft Systems

TP2 Identify the Materials Used in Aircraft Manufacturing

TP3 Discuss Careers Within the Aircraft Manufacturing Industry

TP1 Identify Aircraft Systems

- **AIRCRAFT INSTRUMENT SYSTEMS**
- **AIRFRAME ELECTRICAL SYSTEMS**
- **HYDRAULIC AND PNEUMATIC POWER SYSTEMS**
- **AIRCRAFT LANDING GEAR SYSTEMS**
- **AIRCRAFT FUEL SYSTEMS**



AIRCRAFT INSTRUMENT SYSTEMS

- Instruments that depict airspeed and altitude of aircraft
- Engine operational parameters and electrical system performance
- Microprocessors and other digital electronics

AIRFRAME ELECTRICAL SYSTEMS

- Generate and route electricity to various aircraft components

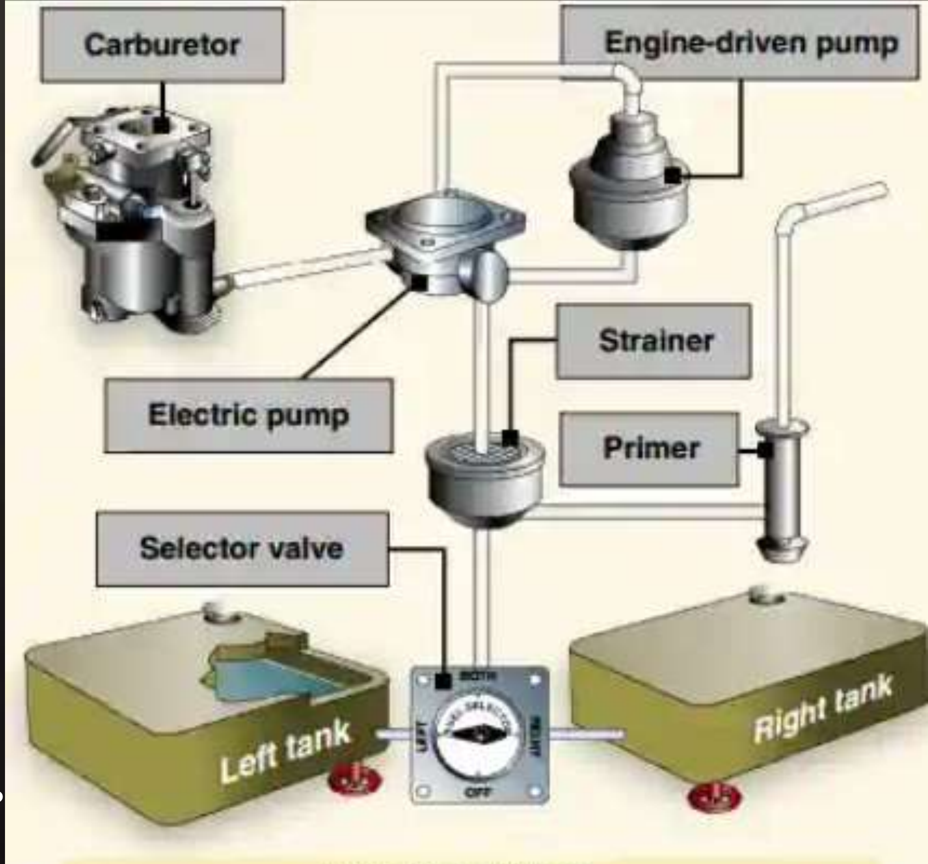
HYDRAULIC AND PNEUMATIC POWER SYSTEMS

- Necessity to operate systems remotely
- Instead of cables or pushrods operating the brakes, hydraulic pressure was used to solve routing problems and multiple forces on the braking surfaces

AIRCRAFT LANDING GEAR SYSTEMS

- Retraction systems, shock absorbing and non-shock

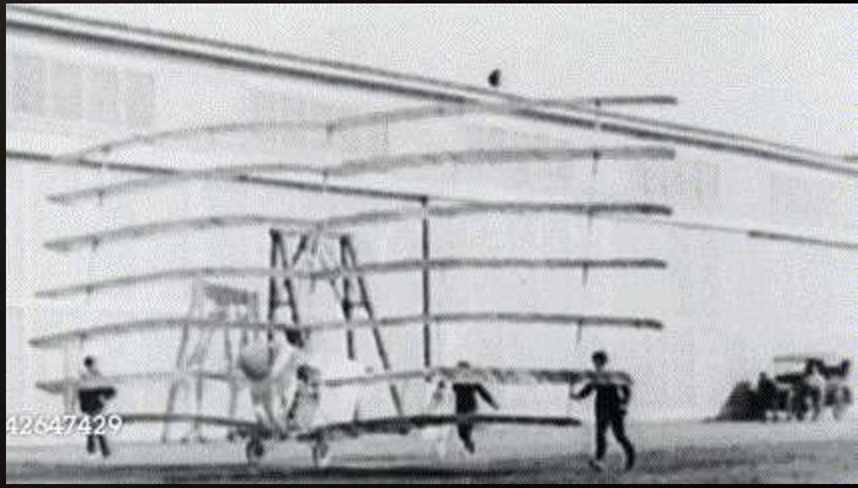
AIRCRAFT FUEL SYSTEMS



Fuel-pump system

- Every fuel system has:
- 1or more fuel tanks
- Tubing to carry the fu the tank(s) to the eng
- Valves to control the flow of fuel
- Provisions for trapping water and contaminants and a method for indicating the fuel quantity

TP2 Identify the Materials Used in Aircraft Manufacturing



- The first aircraft was created using steel, wire, cable, silk and wood
- Metallurgists found that alloying aluminum with other metals resulted in an equivalent to structural steel
- Today, military aircraft

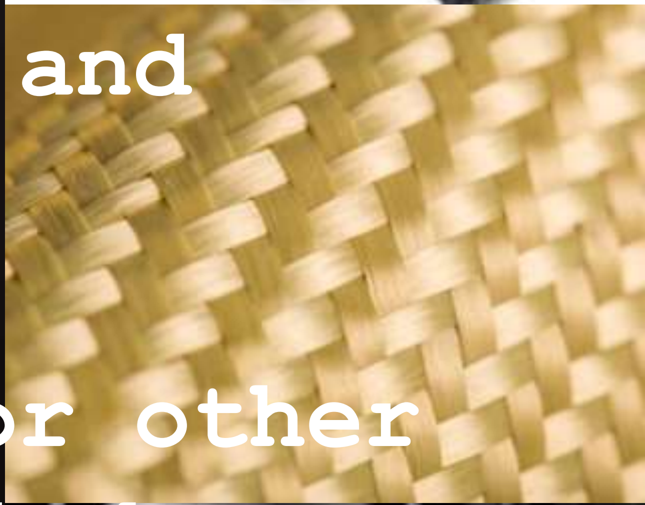
NON-FERROUS METALS

- Aluminum and Its Alloys. Pure aluminum lacks sufficient strength
  - When alloyed, is as strong as steel with 1/3 weight
- Titanium and Its Alloys -lightweight metals with very high strength
- Nickel Alloys
  - Monel works well in gears that require high strength/toughness




1.2 Identify the Materials Used in Aircraft Manufacturing

- COMPOSITE FIBRES
- Graphite Fibres
  - Extremely lightweight, strong, and tough
- Kevlar Fibre
  - Useful where damage from sand or other debris occur – landing gear/behind propellers
- Glass Fibre/Fibreglass
  - Hardens when heated



TP3 Discuss Careers Within the Aircraft Manufacturing Industry

Q1. Share with the class three pieces of interesting information that you did not know about careers in the aircraft manufacturing industry.
Q2. Why did you find this information interesting?
Q3. What was the most interesting career to you?
Q4. What are the primary responsibilities of the career that you found most interesting?